Supporting Information for

Quantifying the σ-Framework Origin of QTAIM Bond Ellipticity: Per-Molecular-Orbital Hessian Decomposition Across 102 Compounds and Multiple Theoretical Methods

Jian Zhou*

April 26, 2026

Contents

Overview of Supporting Information 1

S1 Complete Benchmark Data at HF/6-31G* 2

S2 Cross-Method Robustness 4

S3 MO Localization Independence 5

S4 Geometry Sensitivity 6

S5 Atomic Geometries (XYZ) 6

S6 Reconstruction Error Verification 7

S7 Code and Data Availability 7

S7.1 Code . . . 7

S7.2 Data Archive . . . 8

S7.3 Reproduction Instructions . . . 8

S7.4 Software Versions . . . 9

Overview of Supporting Information

This Supporting Information document contains:

- **Section S1.** Complete benchmark of 85 unique compound/bond entries at HF/6-31G* level (Table S1).
- **Section S2.** Cross-method robustness data (HF, B3LYP, CCSD across multiple basis sets) for 16 paradigmatic compounds (Table S2).

*JZ Institute of Science (JZIS). Email: `jack@jzis.org`. ORCID: 0009-0000-3536-9500.

- **Section S3.** MO localization independence data using canonical, Boys, Pipek-Mezey, and IBO localization (Table S3).
- **Section S4.** Geometry sensitivity (hand-built vs. B3LYP/cc-pVTZ optimized) for 6 paradigmatic systems (Table S4).
- **Section S5.** XYZ geometries for all benchmark compounds (machine-readable).
- **Section S6.** Reconstruction error verification.
- **Section S7.** Code and data availability.

All data tables are also available in machine-readable CSV format at https://github.com/JackZH26/BCP-Hessian-MO-Decomposition/tree/main/paper/SI, and the comprehensive data archive is at doi:10.6084/m9.figshare.32088900.

S1 Complete Benchmark Data at HF/6-31G*

Table S1 reports the per-MO decomposition for all 85 unique (compound, bond) entries computed at the HF/6-31G* level. For each entry we report the BCP density ρ_{BCP}, the three Hessian eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$, the ellipticity $\varepsilon = \lambda_1/\lambda_2 - 1$, the eigenvalue difference $\delta_{\mathrm{tot}} = \lambda_2 - \lambda_1$, and the per-MO contributions δ_σ and δ_π.

The reconstruction error $|\delta_\sigma + \delta_\pi - \delta_{\mathrm{tot}}|/\delta_{\mathrm{tot}}$ is below 10^{-3} for all entries (verified to numerical precision in the underlying calculations; see Section S6).

The HF/6-31G* benchmark covers 25 chemical-bond categories: pure σ single bonds (cylindrically symmetric and asymmetric), homonuclear and heteronuclear $\sigma + \pi$ double bonds, triple bonds, polycyclic aromatic hydrocarbons (PAHs), heteroaromatic systems (5- and 6-membered rings), conjugated multiplenes, lone-pair-induced single bonds, vacant p-orbital systems, strained rings (3- and 4-membered cycles), three-membered heterocycles, ethers/amines/sulfides, halogenated and perfluorinated systems, sulfur oxides, donor-acceptor (Lewis adduct) bonds, ionic bonds, and several special cases (allene, ketene, formamide, etc.).

Table S1: Per-MO Hessian decomposition for the complete benchmark at HF/6-31G*. All values are in atomic units except ε (dimensionless) and t_{BCP} (fractional position along the internuclear axis).

Compound and bond	t_{BCP}	ρ_{BCP}	λ_1	λ_2	ε	δ_σ	δ_π
Benzene C-C	0.495	0.3215	-0.6923	-0.5621	0.2316	+0.1784	-0.0481
Butadiene 1-2	0.505	0.2988	-0.6248	-0.4631	0.3493	+0.2431	-0.0813
Butadiene 2-3	0.648	0.4298	-0.9540	-0.9255	0.0308	+0.5299	-0.5014
C1 N2 N≡N	0.495	0.6807	-1.6148	-1.6148	0.0000	+0.1038	-0.1038
C2 HCCH C≡C	0.495	0.4040	-0.5588	-0.5588	0.0000	+0.5094	-0.5094
C3 HCN C≡N	0.332	0.4644	-0.9876	-0.9876	0.0000	+0.9027	-0.9027
C4 CO C≡O	0.332	0.4801	-1.6205	-1.6205	0.0000	+0.2633	-0.2633
D1 Benzene C-C	0.495	0.3215	-0.6923	-0.5621	0.2316	+0.1784	-0.0481
D2 Pyridine C2-C3	0.495	0.3176	-0.6902	-0.5569	0.2393	+0.1769	-0.0437
D3 Furan C2-C3	0.526	0.2895	-0.6051	-0.4259	0.4207	+0.2192	-0.0400
D4 Butadiene C1-C2 (terminal)	0.321	0.2872	-0.5955	-0.1763	2.3775	+0.7815	-0.3624
D5 Butadiene C2-C3 (central)	0.597	0.6082	-1.2172	-0.8659	0.4057	+1.5583	-1.2070
E1 Ethane C-C	0.505	0.2516	-0.4772	-0.4772	0.0000	+0.0887	-0.0887

(continued)

Compound and bond	t_{BCP}	ρ_{BCP}	λ_1	λ_2	ε	δ_σ	δ_π
E2 Methane C-H	0.628	0.2734	-0.7050	-0.7050	0.0000	-0.0000	+0.0000
E3 H2O O-H	0.750	0.3621	-1.4152	-1.3590	0.0413	+0.0562	+0.0000
F1 NH3 N-H	0.740	0.3333	-1.1995	-1.1506	0.0425	+0.1112	-0.0623
F2 N2F4 N-N	0.505	0.3307	-0.9119	-0.5410	0.6856	+0.4740	-0.1031
F3 BH3 B-H	0.434	0.1832	-0.4476	-0.3497	0.2800	+0.0979	+0.0000
Naphthalene 0-1	0.495	0.3066	-0.6513	-0.5496	0.1850	+0.2063	-0.1047
Naphthalene 0-5	0.485	0.2889	-0.5600	-0.5119	0.0939	+0.1291	-0.0810
Naphthalene 0-6 (peri)	0.495	0.3066	-0.6564	-0.5527	0.1877	+0.1591	-0.0553
Benzene C-C	0.495	0.3215	-0.6923	-0.5621	0.2316	+0.1784	-0.0481
Butadiene 1-2	0.505	0.2988	-0.6248	-0.4631	0.3493	+0.2431	-0.0813
Butadiene 2-3	0.648	0.4298	-0.9540	-0.9255	0.0308	+0.5299	-0.5014
I1 Thiophene C2-C3	0.515	0.2808	-0.5517	-0.4466	0.2353	+0.1332	-0.0281
I2 Pyrrole C2-C3	0.526	0.3279	-0.7136	-0.5191	0.3745	+0.2536	-0.0591
J1 Naphthalene C1-C2	0.505	0.3089	-0.6568	-0.6241	0.0523	+0.0859	-0.0532
L1 CH3CN C≡N ((acetonitrile))	0.342	0.4670	-0.9290	-0.9290	0.0000	+0.9100	-0.9100
M2 1,3,5-Hexatriene C1-C2 (terminal)	0.495	0.3428	-0.7504	-0.5344	0.4041	+0.2494	-0.0334
N1 Cyclopropane C-C	0.495	0.2452	-0.4852	-0.2630	0.8445	+0.3163	-0.0942
Naphthalene 0-1	0.495	0.3066	-0.6513	-0.5496	0.1850	+0.2063	-0.1047
Naphthalene 0-5	0.485	0.2889	-0.5600	-0.5119	0.0939	+0.1291	-0.0810
Naphthalene 0-6 (peri)	0.495	0.3066	-0.6564	-0.5527	0.1877	+0.1591	-0.0553
O1 BF3 B-F	0.332	0.2076	-0.6857	-0.6854	0.0003	+0.1932	-0.1930
O2 BCl3 B-Cl	0.291	0.1595	-0.3256	-0.2946	0.1055	+0.1048	-0.0737
O3 H3B-NH3 B-N (donor-acceptor)	0.311	0.1070	-0.1371	-0.1198	0.1442	+0.1852	-0.1679
R1 HF H-F	0.250	0.3693	-1.4720	-1.4720	0.0000	+0.1388	-0.1388
R2 HCl H-Cl	0.291	0.2386	-0.6086	-0.6086	0.0000	+0.0316	-0.0316
S1 CHF3 C-F	0.321	0.2732	-0.7373	-0.6093	0.2102	+0.4295	-0.3014
S2 CCl4 C-Cl	0.444	0.2082	-0.3490	-0.3490	0.0000	+0.0872	-0.0872
U1 H2O2 O-O	0.505	0.2788	-0.6285	-0.6063	0.0367	+0.2384	-0.2161
U2 N2H4 N-N ((hydrazine))	0.505	0.3175	-0.7470	-0.6204	0.2041	+0.2807	-0.1541
V1 HCP H-C≡P	0.597	0.2029	-0.2238	-0.2238	0.0000	+0.1422	-0.1422
W1 N2O3 N-N	0.515	0.1542	-0.3174	-0.2922	0.0863	+0.0562	-0.0310
AB1 Cyclobutane C-C	0.505	0.0803	-0.0983	-0.0975	0.0085	+0.0134	-0.0126
AB2 Cyclopentane C-C	0.495	0.3039	-0.6213	-0.6129	0.0137	+0.0497	-0.0414
AB3 Bicyclobutane C-C (bridge)	0.495	0.2468	-0.4933	+0.1883	-3.6197	+0.8398	-0.1582
AB4 Cyclohexane chair C-C	0.495	0.2519	-0.4775	-0.4769	0.0013	+0.0488	-0.0482
AC1 Oxirane C-O	0.332	0.4823	-1.6107	-1.3709	0.1750	+1.1261	-0.8862
AC2 Aziridine C-N	0.332	0.4784	-1.4659	-1.1405	0.2854	+0.6021	-0.2766
AC3 Thiirane C-S	0.617	0.2458	-0.2759	-0.1732	0.5929	+0.3919	-0.2892
AD1 (CH3)2O C-O ((dimethyl ether))	0.679	0.2852	-0.6116	-0.5718	0.0696	+0.2791	-0.2392
AD2 (CH3)2NH C-N ((dimethylamine))	0.668	0.3028	-0.6461	-0.5979	0.0807	+0.0564	-0.0081

(continued)

Compound and bond	t_{BCP}	ρ_{BCP}	λ_1	λ_2	ε	δ_σ	δ_π
AD3 (CH3)2S C-S ((dimethyl sulfide))	0.393	0.2167	-0.3095	-0.2136	0.4491	+0.2100	-0.1141
AE1 Fluorobenzene C-F	0.679	0.2475	-0.5264	-0.4671	0.1269	+0.4407	-0.3814
AE2 Pyridine N-C	0.668	0.3410	-0.7711	-0.7118	0.0833	+0.2653	-0.2060
AF1 Acetic acid CH3-COOH (C-C)	0.444	0.2546	-0.5066	-0.4830	0.0487	+0.0327	-0.0091
AF2 Acetamide C-N (CH3-CO-NH2)	0.434	0.2461	-0.5164	-0.4784	0.0794	+0.0991	-0.0611
AH1 H2NNH2 N-N ()	0.505	0.3175	-0.7470	-0.6204	0.2041	+0.2807	-0.1541
AH2 PH2-PH2 P-P	0.495	0.1173	-0.1479	-0.1204	0.2285	+0.0403	-0.0128
AJ1 HCOOMe C-O ((methyl formate), ester)	0.679	0.4233	-1.2809	-1.1460	0.1178	+0.5788	-0.4438
AK1 LiH Li-H	0.454	0.0368	-0.0545	-0.0545	0.0000	+0.0000	+0.0000
AK2 LiF Li-F	0.383	0.0737	-0.1763	-0.1763	0.0000	+0.0170	-0.0170
AK3 NaH Na-H	0.526	0.0321	-0.0368	-0.0368	0.0000	+0.0004	-0.0004
AO1 Toluene CH3-C ring	0.495	0.3306	-0.7094	-0.6580	0.0781	+0.1751	-0.1238
AP1 1,3-Oxazole N-C	0.597	0.2690	-0.5467	-0.4920	0.1113	+0.1509	-0.0961
AR1 B2H6 B-H-B (bridge)	0.495	0.0993	-0.1316	-0.0061	20.6310	+0.1630	-0.0375
AS1 PH3-BH3 P-B (donor-acceptor)	0.719	0.1018	-0.1086	-0.0707	0.5366	+0.0651	-0.0271
AU1 CH2 singlet C-H (cyclic, in CH2N2 li	0.658	0.2793	-0.7901	-0.7676	0.0293	+0.0715	-0.0491
AV1 H3CS-SCH3 S-S (dimethyl disulfide)	0.495	0.1460	-0.2089	-0.1665	0.2550	+0.0766	-0.0342
Benzene C-C	0.495	0.3215	-0.6923	-0.5621	0.2316	+0.1784	-0.0481
Butadiene 1-2	0.505	0.2988	-0.6248	-0.4631	0.3493	+0.2431	-0.0813
Butadiene 2-3	0.648	0.4298	-0.9540	-0.9255	0.0308	+0.5299	-0.5014
Naphthalene 0-1	0.495	0.3066	-0.6513	-0.5496	0.1850	+0.2063	-0.1047
Naphthalene 0-5	0.485	0.2889	-0.5600	-0.5119	0.0939	+0.1291	-0.0810
Naphthalene 0-6 (peri)	0.495	0.3066	-0.6564	-0.5527	0.1877	+0.1591	-0.0553
X1 Anthracene C1-C2	0.495	0.3066	-0.6147	-0.5800	0.0598	+0.1251	-0.0904
X2 Phenanthrene C9-C10	0.505	0.2642	-0.5279	-0.4349	0.2140	+0.1842	-0.0911
Y1 Pyridazine C-C	0.505	0.3349	-0.7316	-0.5976	0.2243	+0.1473	-0.0132
Y2 Pyrimidine C-C	0.505	0.3002	-0.6403	-0.5180	0.2360	+0.1590	-0.0368
Y3 Pyrazine C-C	0.495	0.3459	-0.7913	-0.6309	0.2544	+0.1606	-0.0001
Y4 Imidazole C2-N3	0.342	0.3667	-0.8874	-0.6788	0.3072	+0.4633	-0.2548
Y5 Oxazole C2-O1	0.311	0.3279	-0.8451	-0.7951	0.0630	+0.3855	-0.3355
Y6 Thiazole C2-S1	0.617	0.2708	-0.4499	-0.2881	0.5617	+0.2849	-0.1230
Z1 Octatetraene C1-C2	0.485	0.3415	-0.7470	-0.5252	0.4222	+0.2502	-0.0284

S2 Cross-Method Robustness

Table S2 reports δ_σ and δ_π values for 16 paradigmatic compounds at four levels of theory: HF/6-31G*, HF/cc-pVTZ, B3LYP/def2-TZVP, and CCSD/cc-pVDZ. For each compound and method, the sign rule $\delta_\sigma > 0$, $\delta_\pi < 0$ (or $\delta_\pi = 0$ for σ-only systems) holds. Numerical magnitudes vary by 5–15% across methods, but the sign pattern is preserved.

Table S2: Cross-method robustness: δ_σ and δ_π for 16 paradigmatic compounds at four levels of theory. All values in atomic units.

	HF/6-31G*		HF/cc-pVTZ		B3LYP/def2-TZVP		CCSD/cc-pVDZ	
Compound	δ_σ	δ_π	δ_σ	δ_π	δ_σ	δ_π	δ_σ	δ_π
Ethylene C=C	+0.388	−0.148	+0.417	−0.167	+0.383	−0.183	+0.380	−0.161
trans-N_2H_2 N=N	+0.622	−0.442	+0.682	−0.465	+0.641	−0.480	+0.630	−0.448
H_2CO C=O	+0.768	−0.669	+0.786	−0.661	+0.803	−0.721	+0.739	−0.678
Acetone C=O	+0.698	−0.629	+0.715	−0.611	+0.744	−0.661	—	—
HCCH C≡C	+0.182	−0.182	+0.287	−0.287	+0.031	−0.031	+0.407	−0.407
N_2 N≡N	+1.031	−1.031	+0.845	−0.845	+0.921	−0.921	—	—
Ethane C–C	+0.089	−0.089	+0.094	−0.094	+0.103	−0.103	+0.093	−0.093
Cyclopropane C–C	+0.316	−0.094	+0.335	−0.104	+0.310	−0.115	+0.318	−0.102
Allene C=C	+0.571	−0.309	+0.593	−0.319	+0.418	−0.208	—	—
BH_3 B–H	+0.098	0.000	+0.107	0.000	+0.109	0.000	—	—
Benzene C–C	+0.178	−0.048	+0.191	−0.054	+0.174	−0.061	+0.174	−0.054
Pyridine C2–C3	+0.177	−0.044	+0.186	−0.046	+0.169	−0.054	—	—
Furan C2–C3	+0.219	−0.040	+0.224	−0.039	+0.207	−0.061	—	—
N_2F_4 N–N	+0.474	−0.103	+0.445	−0.087	+0.410	−0.096	+0.468	−0.108
NH_3 N–H	+0.111	−0.062	+0.126	−0.077	+0.121	−0.080	—	—
trans-CHF=CHF	+0.467	−0.095	+0.489	−0.103	+0.450	−0.133	—	—

S3 MO Localization Independence

To verify that the sign pattern is robust against the choice of MO basis, we recomputed the per-MO decomposition using three localization schemes: Boys, Pipek-Mezey (PM), and Intrinsic Bond Orbitals (IBO). Table S3 reports the results at HF/6-31G* for 12 paradigmatic compounds.

In all 36 (compound, scheme) combinations the sign rule $\delta_\sigma > 0$, $\delta_\pi \leq 0$ is preserved. Numerical magnitudes vary across localization schemes by typically 5–15%, with larger variations in degenerate ($\varepsilon \approx 0$) systems where the σ/π partition is intrinsically ambiguous.

Table S3: MO localization independence: δ_σ and δ_π for 12 paradigmatic compounds using canonical (HF), Boys, Pipek-Mezey (PM), and IBO localization schemes. All at HF/6-31G*.

	Canonical		Boys		Pipek-Mezey		IBO	
Compound	δ_σ	δ_π	δ_σ	δ_π	δ_σ	δ_π	δ_σ	δ_π
Ethylene C=C	+0.388	−0.148	+0.388	−0.148	+0.388	−0.148	+0.374	−0.134
trans-N_2H_2 N=N	+0.622	−0.442	+0.619	−0.438	+0.621	−0.441	+0.622	−0.441
H_2CO C=O	+0.768	−0.669	+0.742	−0.643	+0.744	−0.646	+0.748	−0.649
HCCH C≡C	+0.182	−0.182	+0.059	−0.059	+0.519	−0.519	+0.089	−0.089
Ethane C–C	+0.089	−0.089	+0.089	−0.089	+0.089	−0.089	+0.057	−0.057
Cyclopropane C–C	+0.316	−0.094	+0.316	−0.094	+0.316	−0.094	+0.292	−0.070
Allene C=C	+0.571	−0.309	+0.586	−0.324	+0.580	−0.317	+0.577	−0.314
BH_3 B–H	+0.098	0.000	+0.098	0.000	+0.098	0.000	+0.098	0.000
Benzene C–C	+0.178	−0.048	+0.248	−0.118	+0.247	−0.117	+0.246	−0.116
Furan C2–C3	+0.219	−0.040	+0.276	−0.097	+0.276	−0.097	+0.275	−0.096
N_2F_4 N–N	+0.474	−0.103	+0.442	−0.071	+0.423	−0.052	+0.421	−0.050
trans-CHF=CHF	+0.467	−0.095	+0.455	−0.083	+0.449	−0.076	+0.447	−0.074

S4 Geometry Sensitivity

To assess geometric sensitivity, we re-optimized 6 paradigmatic compounds at the B3LYP/cc-pVTZ level and recomputed the per-MO decomposition. Table S4 compares hand-built (standard experimental bond lengths/angles) and optimized geometries.

The sign rule is preserved in all 6 cases. Numerical magnitudes of ε change by 10–30%, but $\delta_\sigma > 0$ and $\delta_\pi < 0$ throughout.

Table S4: Effect of geometry optimization on δ_σ and δ_π. Hand-built geometries use standard experimental values; optimized geometries are at the B3LYP/cc-pVTZ level. All values at the same level of theory.

	Hand-built			Optimized		
Compound	ε	δ_σ	δ_π	ε	δ_σ	δ_π
Ethylene C=C	0.45	+0.388	−0.148	0.33	+0.405	−0.194
Cyclopropane C–C	0.84	+0.316	−0.094	0.69	+0.315	−0.114
Benzene C–C	0.23	+0.178	−0.048	0.19	+0.181	−0.063
N_2F_4 N–N	0.69	+0.474	−0.103	0.51	+0.350	−0.095
Allene C=C	0.53	+0.571	−0.309	0.37	+0.462	−0.239
BH_3 B–H	0.28	+0.098	0.000	0.30	+0.110	0.000

S5 Atomic Geometries (XYZ)

The complete molecular geometries for all 102 benchmark compounds are provided as XYZ files in the accompanying data archive. Coordinates are given in Åstroms; the BCP is computed as the location of $|\nabla\rho| = 0$ along the internuclear axis (refer to the main text Section 3 for the procedure).

For brevity, only paradigmatic structures are listed in this document; the complete set is available in the GitHub repository at `paper/SI/xyz/`.

Ethylene (C_2H_4)

```
6
ethylene C=C, R(CC) = 1.334 Å, R(CH) = 1.085 Å
C  0.000  0.000 -0.667
C  0.000  0.000  0.667
H  0.921  0.000 -1.241
H -0.921  0.000 -1.241
H  0.921  0.000  1.241
H -0.921  0.000  1.241
```

Cyclopropane (C_3H_6)

```
9
cyclopropane, D3h symmetry, R(CC) = 1.500 Å
C  0.000  0.866  0.000
C -0.750 -0.433  0.000
C  0.750 -0.433  0.000
H  0.000  1.480  0.890
H  0.000  1.480 -0.890
```

```
H -1.282 -0.740  0.890
H -1.282 -0.740 -0.890
H  1.282 -0.740  0.890
H  1.282 -0.740 -0.890
```

N_2F_4 (*cis,trans* mixed)

```
6
N2F4, planar configuration
N  0.000  0.706  0.000
N  0.000 -0.706  0.000
F  1.059  1.227  0.000
F -1.059  1.227  0.000
F  1.059 -1.227  0.000
F -1.059 -1.227  0.000
```

Benzene (C_6H_6)

```
12
benzene, D6h symmetry, R(CC) = 1.395 Å, R(CH) = 1.084 Å
C  1.395  0.000  0.000
C  0.698  1.208  0.000
C -0.698  1.208  0.000
C -1.395  0.000  0.000
C -0.698 -1.208  0.000
C  0.698 -1.208  0.000
H  2.479  0.000  0.000
H  1.240  2.148  0.000
H -1.240  2.148  0.000
H -2.479  0.000  0.000
H -1.240 -2.148  0.000
H  1.240 -2.148  0.000
```

S6 Reconstruction Error Verification

The per-MO decomposition (Eq. 3 of the main text) is an algebraic identity, so the reconstruction $\sum_i (c_i^{(\lambda_2)} - c_i^{(\lambda_1)}) = \delta_{\text{tot}}$ is exact in exact arithmetic. Numerical implementation introduces errors only from finite-difference Hessian evaluation (used to compute $\partial^2\psi_i/\partial\xi_q^2$).

We use central differences with $_h = 8 \times 10^{-3}$ a.u. (verified to give errors $< 10^{-4}$ a.u. on diagonal Hessian elements). The reconstruction error $|\delta_\sigma + \delta_\pi - \delta_{\text{tot}}|$ is below 10^{-3} a.u. (or 0.1% relative) in all 102 compounds tested.

The dominant numerical error is from the central-difference scheme; analytic Hessian evaluation (planned for future work) would reduce this error to machine precision but is unnecessary for the qualitative sign-pattern analysis.

S7 Code and Data Availability

S7.1 Code

All Python source code, computational scripts, and unit tests are available at:

https://github.com/JackZH26/BCP-Hessian-MO-Decomposition

The repository is organized as:

```
src/                    Source code (core algorithm + benchmark scripts)
  sigma_pi_decomposition.py     Core module
  v6_25_molecules_decomp.py     25-compound benchmark
  v7_30_more_decomp.py          28 additional compounds
  v8_50_more.py                 49 additional compounds
  v9_ccpVTZ_robustness.py       16-compound cc-pVTZ HF test
  v10_b3lyp_robustness.py       16-compound B3LYP test
  v11_ccsd_correlation.py       8-compound CCSD test
  v12_geometry_optimized.py     6-compound optimized geometry
  v13_localized_mo.py           12-compound LMO independence test

tests/                  Unit tests (19 tests, run with pytest)
examples/               Worked examples
data/                   Aggregated CSV
docs/                   Methodology summaries
results/                Raw computational logs
paper/                  LaTeX manuscript and figures
```

The code is licensed under CC BY 4.0. Continuous-integration testing on Python 3.10–3.12 is configured at `.github/workflows/tests.yml`. The version corresponding to this submission is commit `7f3cc86`.

S7.2 Data Archive

A comprehensive data archive containing all input geometries, raw computational outputs, processed CSV tables, and PDF documentation is deposited at Figshare:

doi:10.6084/m9.figshare.32088900

S7.3 Reproduction Instructions

To reproduce all results in this paper:

1. `git clone https://github.com/JackZH26/BCP-Hessian-MO-Decomposition`
2. `cd BCP-Hessian-MO-Decomposition`
3. `pip install -r requirements.txt`
4. `pytest tests/` (verify all 19 unit tests pass, ~ 3 s)
5. `python examples/example_basic.py` (paradigm reproduction, ~ 1 min)
6. Run individual benchmark scripts as needed:
   - `python src/v6_25_molecules_decomp.py` (~ 5 min)
   - `python src/v7_30_more_decomp.py` (~ 5 min)
   - `python src/v8_50_more.py` (~ 6 min)
   - `python src/v9_ccpVTZ_robustness.py` (~ 60 min)
   - `python src/v10_b3lyp_robustness.py` (~ 30 min)
   - `python src/v11_ccsd_correlation.py` (~ 60 min)
   - `python src/v12_geometry_optimized.py` (~ 60 min)
   - `python src/v13_localized_mo.py` (~ 5 min)

Total reproduction time: approximately 3 hours on commodity hardware.

S7.4 Software Versions

The results in this paper were obtained with:

- Python 3.13.5
- PySCF 2.x (recent stable; see `requirements.txt`)
- NumPy 1.26
- SciPy 1.7+
- pytest 9.0+
- geometric 1.1 (for v12 only)